INTERNSHIP ASSESSMENT REPORT – CIE 2

CHAPTER - 02: ROLES AND RESPONSIBILITIES DURING OJT-1

2.1 Introduction

During my On-the-Job Training (OJT-1), I worked on developing an AI-based Emotion Detection System. The objective of this project was to identify human emotions from facial expressions using artificial intelligence techniques. As part of my internship, I built a web-based application that allows users to upload images and receive emotion predictions such as happy, sad, and angry. I was responsible for backend development using Python and Flask, as well as frontend design using HTML and CSS. For emotion detection, I integrated DeepFace, a pre-trained deep learning library, and used OpenCV for image processing and face detection. This experience enabled me to apply my technical knowledge in a real-world setting and significantly enhanced my practical and development skills.

2.2 Ability to Apply Skills & Technical Knowledge

During my internship, I effectively applied my knowledge of Artificial Intelligence, Machine Learning, and Python programming to develop an Emotion Detection System. For emotion prediction from facial images, I used DeepFace, a pre-trained deep learning library that performs facial analysis and emotion recognition.

I implemented the application using the Flask framework, which helped in building the backend and handling user requests. For basic image handling tasks such as loading and reading images, I used OpenCV. The face detection, preprocessing, and emotion analysis were handled internally by DeepFace.

This project helped me understand how theoretical concepts of AI and Machine Learning can be applied in real-world applications. I was able to integrate multiple technologies and develop a complete working system, which significantly improved my practical and problem-solving skills.

2.3 Intern's Performance on Assigned Tasks and Project

During my internship, I successfully completed all the assigned tasks related to the development of the Emotion Detection System. I was actively involved in designing, developing, and testing the application.

I developed the frontend pages such as login, dashboard, upload, and result display using HTML and CSS. On the backend, I implemented the application logic using Python and Flask, ensuring smooth functionality of the system.

I also worked on integrating DeepFace, a pre-trained model, into the application and performed basic image handling using OpenCV. Additionally, I tested the system with different images to ensure accurate predictions and resolved issues related to routing and form handling.

Overall, I was able to complete the project efficiently and meet the expected requirements, demonstrating good performance throughout the internship.

2.4 Value Addition to the Organization

During my internship, I contributed by developing a functional AI-based Emotion Detection System that demonstrates real-world application of technology. The system can be used to analyze human emotions from facial expressions, which is useful in various fields such as education, customer analysis, and mental health monitoring.

The project showcases how Artificial Intelligence can be integrated with web applications to create meaningful solutions. It can also be further enhanced and scaled for advanced use cases, making it valuable for future development. Through

this work, I was able to create a practical solution that adds value by demonstrating innovation, technical implementation, and real-world applicability.

2.5 Roles and Responsibilities

- Developed the Emotion Detection system using Python and Flask.
- Designed and implemented frontend pages such as login, dashboard, image upload, and result display using HTML and CSS.
- Integrated DeepFace, a pre-trained deep learning library, for emotion prediction.
- Performed basic image handling using OpenCV, while face detection and preprocessing were handled internally by DeepFace.
- Tested the system with different inputs to ensure accurate and reliable performance.
- Debugged errors related to routing, form handling, and prediction results.
- Enhanced the application by adding features like confidence score, suggestions, and history logging.

2.6 Application of Technical Knowledge

During the development of the project, I applied my technical knowledge by implementing a complete workflow for emotion detection. The uploaded images were processed using DeepFace, which internally handles face detection, preprocessing, and emotion analysis. I used OpenCV for basic image loading and handling before passing the images for prediction.

I then passed the uploaded image to DeepFace to generate emotion predictions. Based on the output, I displayed the detected emotion along with a confidence score and relevant suggestions on the web interface.

This practical implementation helped me understand how theoretical concepts are executed step-by-step in a real-world system.

2.7 Learning Outcomes

During my internship, I gained valuable practical knowledge in developing real-world applications using Artificial Intelligence and web technologies. I learned how to integrate a deep learning model with a web-based system and understood the complete workflow of an AI application.

I improved my skills in Python programming and the Flask framework, and gained experience in image handling using OpenCV. I also worked with DeepFace for emotion detection. Additionally, I learned error handling and debugging, which enhanced my problem-solving skills and confidence in working on real-time projects.

2.8 Personal Reflection

During my internship, I gained valuable hands-on experience in developing an AI-based application. Initially, I faced difficulties in integrating the model and handling image processing tasks, but through continuous practice and debugging, I was able to overcome these challenges.

This project helped me improve my technical knowledge, problem-solving skills, and confidence in working on real-time systems. I learned how to apply theoretical concepts in a practical environment and understood the importance of proper project structure and testing.

2.9 Conclusion

In conclusion, the Emotion Detection System developed during my internship successfully demonstrates the application of Artificial Intelligence and web technologies in a real-world scenario. The project was completed by integrating a deep learning model with a user-friendly web interface, enabling accurate emotion prediction from facial images. I was able to apply my technical knowledge, complete assigned tasks effectively, and develop a solution that has practical value.

2.11 USE CASE: EMOTION DETECTION SYSTEM

2.11.1 Introduction

This use case describes the application of an Emotion Detection System that identifies human emotions from facial expressions using Artificial Intelligence. The system allows users to upload an image, which is then processed to detect emotions such as happy, sad, angry, and neutral.

This system can be used in various real-world scenarios such as smart classrooms to monitor student engagement, customer behaviour analysis in businesses, and mental health monitoring systems.

2.11.2 Problem Statement

In many real-world situations, it is difficult to accurately understand a person's emotional state based only on observation. In environments such as classrooms or customer service, it becomes challenging to monitor engagement, satisfaction, or reactions continuously.

Manual observation is time-consuming and may not always be reliable. Therefore, there is a need for an automated system that can detect and analyze human emotions efficiently using facial expressions.

2.11.3 Proposed Solution

To address the above problem, an AI-based Emotion Detection System is developed. The system allows users to upload facial images, which are then processed to detect emotions using a pre-trained deep learning model.

The application uses OpenCV for basic image handling. The uploaded image is then processed using DeepFace, which internally performs face detection,

preprocessing, and emotion prediction. The result is displayed along with a confidence score and suggestions on the web interface. This solution provides a simple, efficient, and automated way to analyze human emotions in real-time scenarios.

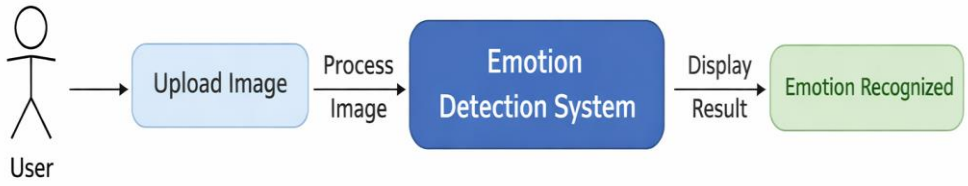
2.11.4 Workflow of the Emotion Detection System

1. User uploads an image through the web interface.
2. The image is read and processed using OpenCV.
3. The system detects the face region in the image.
4. The detected face is extracted and prepared for analysis.
5. The face image is passed to the DeepFace library.
6. DeepFace analyzes the image and predicts the emotion.
7. The system generates a confidence score for the detected emotion.
8. The final emotion and confidence score are displayed to the user.

2.11.5 Use Case Diagram

The following diagram represents the interaction between the user and the Emotion Detection System:

Figure 2.1: Login Page of Emotion Detection System



2.11.6 Output Screenshots

Figure 2.2: Login Page of Emotion Detection System

Figure 2.3: Dashboard Interface

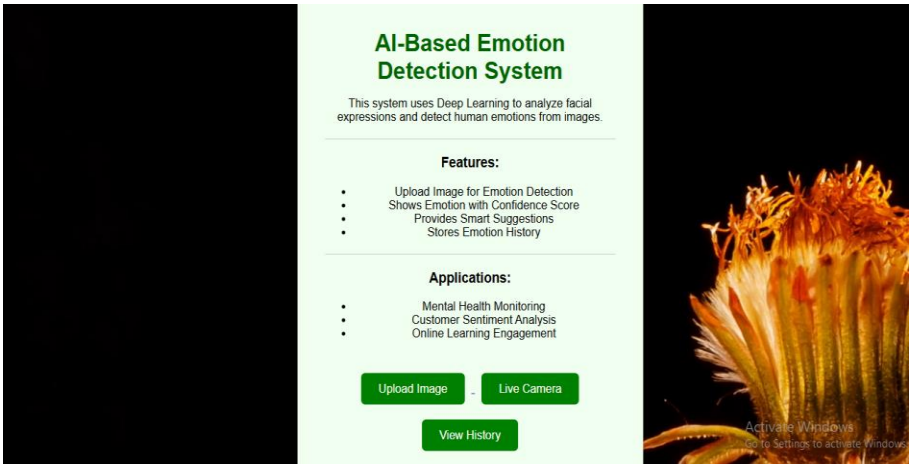


Figure 2.4: Image Upload Page & Result Page

2.11.7 Benefits of the System

- Provides an automated way to detect human emotions accurately.
- Useful in real-world applications such as education, healthcare, and customer analysis.
- Helps in understanding user behavior and emotional responses.
- Improves human-computer interaction by making systems more intelligent.
- Can be extended and scaled for advanced AI-based applications.

2.11.8 Conclusion

The Emotion Detection System use case demonstrates how Artificial Intelligence can be applied to solve real-world problems. The system provides an efficient and user-friendly solution for emotion analysis using facial expressions. Overall, it highlights the practical implementation and usefulness of AI in different domains.